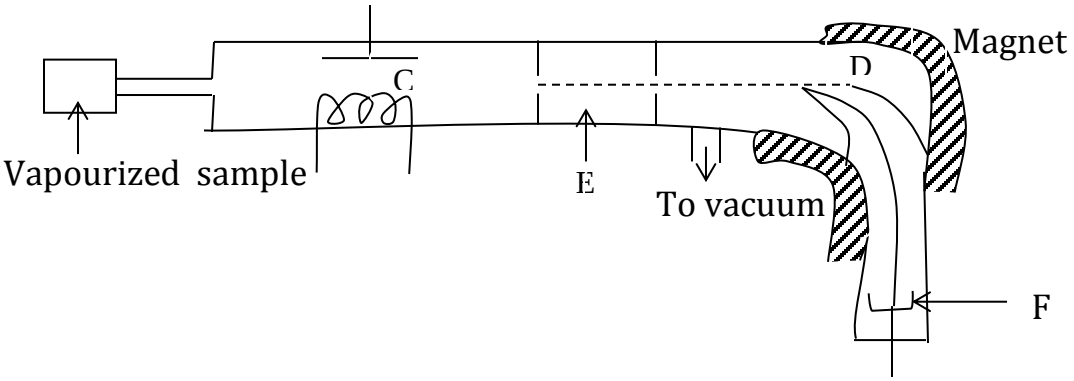
1. (a). Write **the electronic configuration** of element **Gallium** (Ga). (0 ½ marks)

(b). Natural Gallium consists of isotopes ^{69}**Ga** and ^{71} **Ga** in atomic ratio **3:2**.The relative isotopic masses of ^{69}**Ga** and ^{71} **Ga** are **68.9** and **70.9** respectively.

(i). Calculate the approximate relative atomic mass of Gallium. (01½ marks)

(ii). Explain how the mass spectrum can be used to obtain the relative atomic mass of Gallium. (09 marks)

(c). (i). The figure **1.0** below represents a mass spectrometer.



Name and state the functions of the following parts. (04 marks)

C ……

D ……..

E ……..

F ……..

(ii).Explain why the values of **relative atomic masses** have **no units**. (01 mark)

(iii).State **two limitations** of using **mass spectrum** for the determination of relative atomic mass. (02 marks)

(d) Determine the co-ordination numbers and oxidation states of the central metal atom/ion in the following complexes. (02 marks)

Complex	Co-ordination NO:	Oxidation states
$[Fe(CN)_6^{2-}]$		
$[Cr(H_2O)_6Cl_3]$		

2. **10.0cm³** of a hydrocarbon **P**, [C_xH_y] was exploded in **90.0cm³** of oxygen gas. On cooling to room temperature, the residual gases occupied **70.0cm³**.When the residual gases were passed via potassium hydroxide solution, the volume was reduced to **40.0cm³**.
   (a) (i). Write equation for the reaction between **P** and **oxygen gas**. (01 mark)
   (ii).Determine the molecular formula of hydrocarbon **P**. (03 marks)
   (iii).Write the structural formula and name of hydrocarbon P. (01 mark)
   (iv).Write a balanced equation for the combustion of hydrocarbon **P** and oxygen gas. (01½ marks)
   (b)Write equation to show how hydrocarbon **P** can be prepared from a named alcohol. (03 marks)
   (c) Discuss the following types of reaction
   i. Nucleophilic substitution reaction (03 marks)
   ii. Electrophilic addition reaction (03 marks)
   iii. Elimination reaction (03 marks)
   (d)When propene is reacted with bromine gas in presence of tetrachloromethane, 1, 2-dibromopropane is formed but if propene is reacted with bromine gas in presence of sodium chloride solution, 1-bromo-2-chloropropane is formed. Explain the observations. (01½ marks)

3. (a). Explain the variation of ionization energy down group (II) elements. (02½ marks)
   (b). The table below shows the elements in period 2 of the periodic table.

Elements	$_3Li$	$_4Be$	$_5B$	$_6C$	$_7N$	$_8O$	$_9F$	$_{10}Ne$
1st I.E	520	899	801	1086	1403	1310	1681	2080

   i. Plot a graph of first ionization energy against atomic number. (02½ marks)
   ii. Explain the shape of the graph. (06½ marks)

(c). Li, Na, and K belong to group (I) metal elements of the periodic table. Describe the reactions of the elements with oxygen gas.

(08½ marks)

4. (a). (i). State the difference between ideal and real gas. (02 marks)

(ii). State two assumptions of the kinetic theory for ideal gases.

(02 marks)

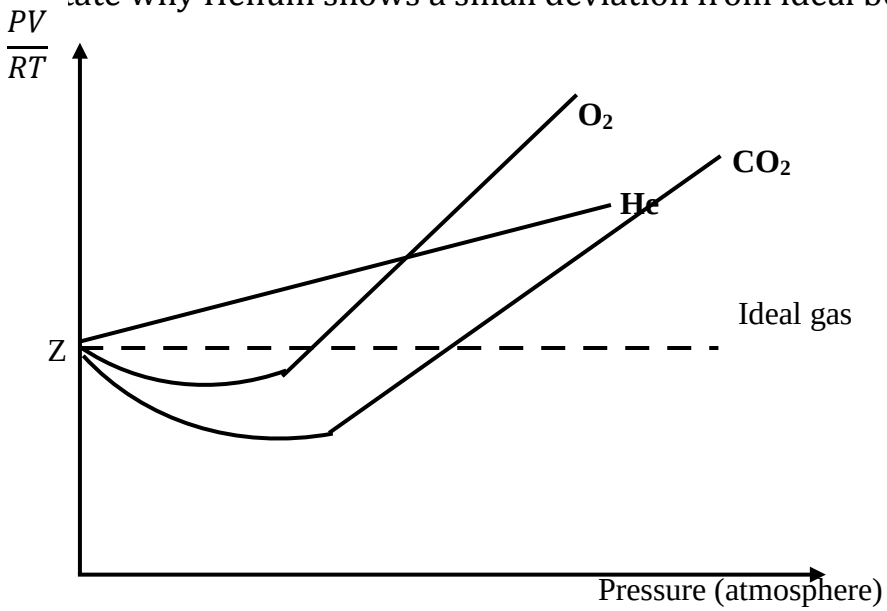
(b). Explain how liquidifaction of a gas can be affected by:

i. Temperature. (02½ marks)

ii. Pressure. (02½ marks)

(c).The curve below shows deviations of some gases from ideal behavior.

i State why Helium shows a small deviation from ideal behavior



compared to other gases. (01½ marks)

ii. Compare the deviation of oxygen and carbon dioxide gas from ideal behavior. (02½ marks)

iii. Briefly explain why at point **Z**, oxygen, helium and carbon dioxide have the same pressure with the ideal gas. (02 marks)

(d). A gas **Q** contains **30.43%** nitrogen and the rest being oxygen gas. **0.23g** of **Q** occupied **154.11cm³** at **150°C** and **840mmHg**.

Determine the:

i. Empirical formula of **gas Q**. (02 marks)

ii. Molecular formula of **gas Q**. (03 marks)

[**1 mole** of a gas occupies **24dm³** at a temperature of **25°C** and pressure of **760mmHg**]

(e).When a mixture of **Q** and oxygen was bubbled through water, a compound **Y** was formed.

i. Identify **Y**.

ii. Write equation for the reaction leading to the formation of **Y**.

iii. Briefly describe a test(s) that can be carried out to identify **Y**.

5. When **8.8g** of hydrocarbon, **Z** was burnt in excess air, **14.4g** of steam and **13.44dm^3** of carbon dioxide gas were obtained at s.t.p.
[1 mole of a gas at s.t.p occupies 22.4dm^3]

a) Determine the empirical formula of hydrocarbon, **Z**. (03½ marks)

b) The vapour density of hydrocarbon, Z is **22**.Write the name of hydrocarbon **Z** after determining the molecular formula. (02½ marks)

c) (i). Write equation indicating conditions to show how hydrocarbon **Z** can be synthesized from a named alcohol. (03 marks)

(ii).Indicate the mechanism for the first stage of the reaction. (03½ marks)

d) Name the reagent used to differentiate the following compounds and in each case state what is observed and write the equation of reaction where possible.

i. $CH_3HC{=}CH_2$ and $CH_3CH_2CH_3$ (03 marks)

ii. Ca^{2+} and Mg^{2+} (03 marks)

e) Explain why dehydration of butan-1-ol and butan-2-ol forms but-2-ene as the major product but not but-1-ene. (01½ marks)